

# Anton Kuttyrev, PhD

Lecturer in Exploration Geology, Cardiff University Kazakhstan, Astana, Kazakhstan

ORCID 0000-0001-5933-6455 · [www.antonkuttyrev.com](http://www.antonkuttyrev.com) · [Google Scholar](#) · [ResearchGate](#)

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## Research focus

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Petrology and mineralogy of mafic–ultramafic and island-arc systems; melt–rock and fluid–rock interaction; origin of chromitites; PGE mineralogy from hand sample to nanoscale, including alloys and PGE oxides; PGE/PGM cycling and mobility in melts and fluids in ultramafic rocks and layered intrusions; oxygen isotopes as tracers of crustal recycling and early Earth surface environments; critical minerals in the context of the energy transition.

## Education

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- 2020      **PhD in Geology and Mineral Sciences.**  
Defended in 2019 and awarded in 2020 through the Dissertation Council at the V.S. Sobolev Institute of Geology and Mineralogy (SB RAS); research carried out at the Institute of Volcanology and Seismology (RAS), Petropavlovsk-Kamchatsky, Russia.
- 2017      **Specialist degree with Distinction in Applied Geology**, Saint Petersburg Mining University, Russia.  
Five-year full-time programme, broadly equivalent to a Master of Science.

## Academic appointments

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- 2026–present      **Lecturer in Exploration Geology**, Cardiff University Kazakhstan, Astana, Kazakhstan.  
Teaching, programme leadership, curriculum development and research in economic and exploration geology, including layered intrusions, chromitites, platinum-group elements, critical minerals and isotope geochemistry. Lead the Exploration Geology programme on the Kazakhstan side and coordinate its local delivery; organising the programme’s first fieldwork and developing a new field course from scratch.
- 2025–2026      **Postdoctoral Research Fellow**, School of Geosciences, University of the Witwatersrand, Johannesburg, South Africa.  
Critical minerals in layered mafic–ultramafic intrusions and Bushveld chromitites within the DSTI CIMERA industry–public consortium, with a focus on Ni-Cu-PGE systems, chromitites and the petrogenesis of the Critical Zone.
- 2023–2025      **Research Associate**, School of Earth and Environmental Sciences, Cardiff University, Wales, UK.  
Critical-mineral and PGE geochemistry in layered intrusions and related volcanic rocks within the Horizon Europe SEMACRET consortium on energy-transition mineral systems, in an interdisciplinary team spanning geology, geophysics and social sciences.
- 2022–2023      **Fulbright Visiting Scholar**, School of Earth Sciences, University of Oregon, USA.  
Early Archean climate from triple oxygen and hydrogen isotopes in 3.8 Ga metalavas of the Saglek-Hebron terrane, including development of isotope fractionation models.
- 2020–2022      **Research Associate**, Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia.  
PGE mineralization and hydrothermal processes in subduction-related magmatic systems of Kamchatka.

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| 2017–2020 | <b>PhD candidate</b> , Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia.<br>Geology and platinum mineralization of concentrically zoned dunite–clinopyroxenite–gabbro complexes of the Tamanvayam and Epilchik groups.                                                                                                                     |
| 2012–2017 | <b>Engineer (0.5 FTE)</b> , Karpinsky Russian Geological Institute, Saint Petersburg, Russia.<br>Regional geology and petrology, including petrography and geochemical data for igneous and metamorphic rocks; work on rock collections of the Academician F.N. Chernyshev Central Geological Research Museum and preparation of a field rock identification guide. |

### Industry appointments

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| 2018      | <b>Geologist (summer fieldwork)</b> , Kirganik Ltd. (Amur Minerals subsidiary), Kamchatka, Russia.<br>Exploration for porphyry Cu and epithermal Au systems: field mapping, alteration and mineralization logging, geochemical sampling. |
| 2016–2017 | <b>Geologist (summer field seasons)</b> , Kamp Ltd., Kamchatka, Russia.<br>Junior exploration for PGE mineralization: prospecting, field mapping, heavy-mineral sampling, rock and soil geochemical sampling.                            |
| 2015      | <b>Field technician (summer fieldwork)</b> , Kamchatgeologia, Kamchatka, Russia.<br>Regional geological mapping at 1:200,000 scale: traverses, outcrop description and systematic sampling for petrography and geochemistry.             |

### Teaching experience

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| 2026–present | <b>Cardiff University Kazakhstan</b> . World of Dynamic Environments (EA3504), a Year 1 module on dynamic Earth systems taught through geological and environmental case studies; Earth Materials and fieldwork from this academic year. Designing a new field course from scratch, including its academic structure, field activities and teaching materials. |
| 2025–2026    | <b>University of the Witwatersrand</b> . Five-lecture module on magmatic mineral deposits for honours students; demonstrator on the honours field trip (Northern Bushveld).                                                                                                                                                                                    |
| 2023–2025    | <b>Cardiff University</b> . Practical sessions for the Mineral Deposits course; guest lectures on mineral deposits; demonstrator on field courses in geological mapping, exploration geology and environmental geoscience (Cyprus, Cornwall, South Wales).                                                                                                     |
| 2022–2023    | <b>University of Oregon</b> . Demonstrator on petrology field courses; guest lectures in petrology and geohazards.                                                                                                                                                                                                                                             |

### Formal teaching training

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| 2024 | Teaching to Improve the Quality of Student Learning, Cardiff University. |
| 2024 | Lecturing Skills, Cardiff University.                                    |

### Student supervision

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| 2026–present | <b>Co-supervisor</b> (with Prof. Roger Gibson) of the project “Calc-silicate xenoliths in the Lower Zone of the Bushveld Complex and their implications for platinum-group element mineralization” (student: Mensacia Phaswane). |
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### Additional professional training

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| 2026 | PetThermoTools virtual workshop.                                         |
| 2025 | Reading the Rocks, Polokwane, South Africa.                              |
| 2025 | Resource Geology Workshop, GeoCongress 2025, Bloemfontein, South Africa. |
| 2024 | MDSG workshop “Magmatic Ni-Cu Mineral Systems in Europe”, Edinburgh, UK. |

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| 2021 | TESCAN electron microscopy course, Saint Petersburg, Russia. |
| 2018 | Gold Short Course, Prague, Czech Republic.                   |

### Analytical and modelling skills

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**Hands-on:** SEM-EDS, Raman spectroscopy, field-portable XRF, clean-lab chemistry and mass spectrometry for Rb-Sr, Sm-Nd and triple-oxygen isotopes (laser fluorination line), LA-ICP-MS.

**Data reduction and interpretation:** EBSD, X-ray diffraction, synchrotron micro-XRF/XAS, FIB-TEM, EPMA, atom-probe tomography.

**Modelling and data processing:** Python and spreadsheet-based workflows; MELTS- and MAGEMin-based thermodynamic modelling; Monte Carlo modelling of triple-oxygen isotope exchange; mass-balance and mixing models.

### Scientific service

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| 2024–2025 | <b>Coordinator</b> , Minerals and Energy Research Centre, Cardiff University.                                                                                                                                                                                                                                                              |
| 2023      | <b>Organizing committee member</b> , 14th International Platinum Symposium, Cardiff, UK.                                                                                                                                                                                                                                                   |
| Reviewer  | for <i>Ore Geology Reviews</i> , <i>Mineralium Deposita</i> , <i>Nature Communications</i> , <i>Chemical Geology</i> , <i>Journal of Petrology</i> , <i>Australian Journal of Earth Sciences</i> , <i>Contributions to Mineralogy and Petrology</i> , <i>Journal of the Geological Society</i> , <i>Lithos</i> and <i>Marine Geology</i> . |

### Competitively awarded funding

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*Amounts converted from foreign currencies to EUR; approximate.*

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| 2024      | <b>AuScope Access Program</b> (Curtin University), in-kind analytical access: “Silicate–oxide–platinum group minerals intergrowths: implications for unknown aspects of low-T PGE behaviour”. Approx. EUR 10,700.            |
| 2022–2023 | <b>Fulbright Scholarship</b> (Fulbright Visiting Scholar). Approx. EUR 26,600.                                                                                                                                               |
| 2020–2022 | <b>RFBR Project No. 20-35-70008</b> (project executor): “Formation and alteration of minerals in active volcanic hydrothermal systems (Kamchatka)”. Approx. EUR 114,000.                                                     |
| 2020–2021 | <b>RFBR Project No. 20-05-00290</b> (principal investigator): “Distribution of Ni-Fe-Co intermetallic compounds and platinum-group minerals in mafic and ultramafic rocks of various formational types”. Approx. EUR 21,500. |
| 2018–2020 | <b>RSCI Project No. 16-17-10145</b> (project executor): “Melt–fluid evolution, magmatic immiscibility, and metal budgeting in subduction-related magmas”. Approx. EUR 171,000.                                               |
| 2018–2019 | <b>RFBR Project No. 18-35-00126</b> (principal investigator): “Typomorphism and formation conditions of PGE minerals in concentrically zoned ultramafic massifs (Epilchik area, Koryak-Kamchatka belt)”. Approx. EUR 9,700.  |
| Total     | <b>Total research funding received: approx. EUR 353,600.</b>                                                                                                                                                                 |

### Field work

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| 2025–present | Mapping and sampling of the Critical Zone across the western and eastern Bushveld Complex, South Africa (one month of fieldwork plus regular mine visits). |
| 2023–2025    | Fieldwork at the Rhiw layered intrusion, Wales, UK (8 days in total).                                                                                      |
| 2021         | Principal investigator and organizer of a helicopter-supported expedition (12 participants) to the Tumrok Range arc picrites, Kamchatka (3 weeks).         |
| 2020         | Sampling of alluvial awaruite in the Kamchatsky Mys ophiolite, Kamchatka (2 weeks).                                                                        |
| 2020         | Documenting cooling lava flows and caves of the 2012–13 Tolbachik Fissure Eruption (2 weeks); observations of Klyuchevskoy volcano.                        |

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| 2019      | Underground and field sampling of mineralized intrusions and related flood basalts, Norilsk (2 weeks).                                    |
| 2018      | Documenting lava temperatures in the caves of the 2012–13 Tolbachik Fissure Eruption (2 weeks); visual observations of Shiveluch volcano. |
| 2018      | Fieldwork on the Kamchatsky Mys ophiolite (2 weeks).                                                                                      |
| 2018      | Prospecting at the Kirganik Cu porphyry and epithermal Au occurrence (1 month).                                                           |
| 2016–2017 | Prospecting for lode and placer PGE mineralization, Epilchik and Matysken areas, Koryak Highlands (5 months in total).                    |
| 2015      | Geological mapping at 1:200,000 scale, Koryak Highlands (3 months).                                                                       |

### Professional societies

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| 2023–present | Mineralogical Society of Great Britain and Ireland.                                                                   |
| 2021–present | Society of Economic Geologists (SEG).                                                                                 |
| 2017–present | Society of Geology Applied to Mineral Deposits (SGA). 2026–present – Scientific advisor of Central Asian SGA Chapter. |
| 2014–present | Russian Mineralogical Society; 2021–2022 – Head of the Kamchatka Department                                           |

### Awards

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| 2023 | Minerals Young Investigator Award.                 |
| 2021 | Marie Skłodowska-Curie Actions Seal of Excellence. |

### Public outreach

School visits in the UK and Russia for ages 8 to 16: interactive lectures on volcanoes, minerals and rocks, the societal impact of geology, and career perspectives of a geoscientist.

### Publications

#### Peer-reviewed articles (35)

1. **Kutyrev, A.**, Filimonova, O., Saxey, D.W., Vymazalová, A., Evans, K., Kamenetsky, V.S., Wermeille, D., Fougereuse, D., Reddy, S.M., 2026. Formation of natural IrO<sub>2</sub> through Os-Ir-Ru alloy oxidation. *Chemical Geology*, 123167. <https://doi.org/10.1016/j.chemgeo.2025.123167>
2. Eslami, A., Chardelin, M., O'Driscoll, B., Malvoisin, B., Evans, K., Grieco, G., Bussolesi, M., & **Kutyrev, A.** 2026. Chromitite-mafic dike interfaces as archives of syn-kinematic fluid-rock interaction, chromium mobility and Ca-metasomatism, Nain Ophiolite, Central Iran. *Journal of Petrology*, egag076. <https://doi.org/10.1093/petrology/egag076>
3. Rhoads, E.A., **Kutyrev, A.**, Bindeman, I.N., Lynn, K.J., Trusdell, F.A., Downs, D.T., Edwards, H.R., Cook, G.W., Day, J.M.D., 2025. Rhenium-osmium and oxygen isotope homogeneity during the 2022 Mauna Loa eruption and implications for basaltic magma storage. *Bulletin of Volcanology* 87. <https://doi.org/10.1007/s00445-025-01825-0>
4. Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Barnes, S.J., Stepanov, A.S., Konstantinov, S.E., Baykov, N.I., Kryazhev, S.G., Belousov, I.A., Zhukova, I.A., Scherbakov, V.D., Kalugin, V.M., Ivanov, A.V., Bryanskiy, N.V., Park, J.W., 2025. Erratic behavior of platinum-group elements along an olivine-Cr-spinel cotectic in sulfide-undersaturated basaltic magma: roles of Cr-spinel and alloys. *Scientific Reports* 15, 14366. <https://doi.org/10.1038/s41598-025-97281-z>
5. Zhitova, E.S., Nuzhdaev, A.A., Davydova, V.O., Sheveleva, R.M., Zhegunov, P.S., Kuznetsov, R.A., **Kutyrev, A.V.**, Khokhlova, M.A., Vlasenko, N.S., 2025. Calcareous skarn-like mineral paragenesis from unaltered basalt of the Alaid volcano (Kuril-Kamchatka island arc). *Minerals* 15. <https://doi.org/10.3390/min15030237>

6. **Kutyrev, A.**, Bindeman, I.N., O'Neil, J., Rizo, H., 2024. Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada: Implications for moderately low- $\delta^{18}\text{O}$  Eoarchean ocean. *Chemical Geology* 670. <https://doi.org/10.1016/j.chemgeo.2024.122378>
7. **Kutyrev, A.**, Kamenetsky, V.S., Chayka, I.F., Nekrylov, N.A., Kryuchkova, L., Shilovskikh, V.V., Kontonikas-Charos, A., Krashenninnikov, S.P., Sapegina, A., Perchuk, A.L., 2024. Metamorphic origin of large nuggets of platinum-group metals: evidence from multiphase inclusions in Os-Ir-Ru alloys from the Adamsfield placer, Tasmania. *Journal of Petrology*. <https://doi.org/10.1093/petrology/egae086>
8. Zhegunov, P.S., **Kutyrev, A.V.**, Zhitova, E.S., Moskaleva, S.V., Schweigert, P.E., 2024. First data on the mineralogy of the Evevpena epithermal silver-gold ore occurrence, Kamchatka, Russia. *Journal of Volcanology and Seismology* 18, 149-164. <https://doi.org/10.1134/S0742046324700489>
9. Stepanov, S., Palamarchuk, R., **Kutyrev, A.**, Lepekhina, E., Sharpenok, L., Shagalov, E., Minervina, E., 2024. Nature of perovskite mineralization of silicate-carbonate veins in the margins of Kusinsko-Kopanskaya layered intrusion (South Urals, Russia). *Minerals* 14, 478. <https://doi.org/10.3390/min14050478>
10. **Kutyrev, A.**, McDonald, A., Tamura, N., Cabri, L.J., Tolstykh, N., Pekov, I., 2023. Sidorovite,  $\text{PtFe}_3$ , a new mineral from the Snegovaya River placer, Koryak Highlands, Russia. *The Canadian Journal of Mineralogy and Petrology* 61(5), 1021-1030. <https://doi.org/10.3749/2200072>
11. **Kutyrev, A.**, Kamenetsky, V.S., Kontonikas-Charos, A., Savelyev, D.P., Yakich, T.Y., Belousov, I.A., Sandimirova, E.I., Moskaleva, S.V., 2023. Behavior of platinum-group elements during hydrous metamorphism: constraints from awaruite ( $\text{Ni}_3\text{Fe}$ ) mineralization. *Lithosphere*, 2023(1). [https://doi.org/10.2113/2023/lithosphere\\_2023\\_126](https://doi.org/10.2113/2023/lithosphere_2023_126)
12. Chayka, I.F., Izokh, A.E., Kamenetsky, V.S., Sokol, E.V., Lobastov, B.M., Kontonikas-Charos, A., Zelenski, M.E., **Kutyrev, A.V.**, Sluzhenikin, S.F., Zhitova, L.M., Shvedov, G.I., Shevko, A.Y., Gora, M.P., 2023. Origin of chromitites in the Norilsk-1 intrusion (Siberian LIP) triggered by assimilation of argillaceous rocks by Cr-rich basic magma. *Lithos* 454-455. <https://doi.org/10.1016/j.lithos.2023.107254>
13. Eslami, A., Grieco, G., Bussolesi, M., Ichiyama, Y., Lenaz, D., Skogby, H., **Kutyrev, A.V.**, Cavallo, A., Khedr, M.Z., 2023. Co-occurrence of compositionally variable chromitites in the Sabzevar ophiolite, NE Iran. *Lithos* 446-447. <https://doi.org/10.1016/j.lithos.2023.107133>
14. Stepanov, S.Y., Palamarchuk, R.S., **Kutyrev, A.V.**, Shilovskikh, V.V., Petrov, S.V., 2023. Mineralogy and geochemistry of platinum-group elements in the zoned mafic-ultramafic intrusions of the Uralian Platinum belt, Russia. *Journal of Geochemical Exploration* 246. <https://doi.org/10.1016/j.gexplo.2023.107155>
15. Chayka, I.F., Baykov, N.I., Kamenetsky, V.S., **Kutyrev, A.V.**, Pushkarev, E.V., Abersteiner, A., Shcherbakov, V.D., 2023. Volcano-plutonic complex of the Tumrok Range (Eastern Kamchatka): an example of the Ural-Alaskan type intrusion and related volcanic series. *Minerals* 13, 126. <https://doi.org/10.3390/min13010126>
16. Yakubovich, O., **Kutyrev, A.**, Sidorov, E., Travin, A., 2022.  $^{190}\text{Pt}$ - $^4\text{He}$  dating of platinum mineralization in Ural-Alaskan-type complexes in the Kamchatka region: evidence for remobilization of platinum-group elements. *Mineralium Deposita*. <https://doi.org/10.1007/s00126-022-01103-5>
17. Bukhanova, D.S., **Kutyrev, A.V.**, Sidorov, E.G., Chubarov, V.M., 2022. First finding of platinum group minerals at the Malmyzh porphyry gold-copper deposit, Khabarovsk Krai, Russia. *Geology of Ore Deposits* 63, 675-681. <https://doi.org/10.1134/s1075701521070035>
18. Pekov, I.V., Zubkova, N.V., Agakhanov, A.A., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Sidorov, E.G., **Kutyrev, A.V.**, Pushcharovsky, D.Y., 2022. New arsenate minerals from the Arsenatnaya fumarole, Tolbachik volcano, Kamchatka, Russia. XIX. Axelite,  $\text{Na}_{14}\text{Cu}_7(\text{AsO}_4)_8\text{F}_2\text{Cl}_2$ . *Mineralogical Magazine*. <https://doi.org/10.1180/mgm.2022.120>
19. Pekov, I.V., Zubkova, N.V., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Agakhanov, A.A., Turchkova, A.G., Sidorov, E.G., **Kutyrev, A.V.**, Blatov, V.A., Pushcharovsky, D.Y., 2022. Nishanbaevite,  $\text{KAl}_2\text{O}(\text{AsO}_4)(\text{SO}_4)$ , a new As/S-ordered arsenate-sulfate mineral of fumarolic origin. *Mineralogy and Petrology*. <https://doi.org/10.1007/s00710-022-00803-0>

20. Cui, M.-M., Su, B.-X., Wang, J., Wu, Y., **Kutyrev, A.**, 2022. Early Paleozoic subduction fingerprints of the Paleo-Asian ocean in easternmost Central Asian Orogenic Belt (CAOB): Identification of the oldest Alaskan-type complex in the CAOB. *GSA Bulletin*. <https://doi.org/10.1130/B36307.1>
21. Nekrylov, N., Kamenetsky, V.S., Savelyev, D.P., Gorbach, N.V., Kontonikas-Charos, A., Palesskii, S.V., Shcherbakov, V.D., **Kutyrev, A.V.**, Savelyeva, O.L., Korneeva, A.A., Kozmenko, O.A., Zelenski, M.E., 2022. Platinum-group elements in Late Quaternary high-Mg basalts of eastern Kamchatka: Evidence for minor cryptic sulfide fractionation in primitive arc magmas. *Lithos*. <https://doi.org/10.1016/j.lithos.2022.106608>
22. **Kutyrev, A.V.**, Kamenetsky, V.S., Park, J.-W., Maas, R., Demonterova, E.I., Antsiferova, T.N., Ivanov, A.V., Hwang, J., Abersteiner, A., Ozerov, A.Y., 2021. Primitive high-K intraoceanic arc magmas of Eastern Kamchatka: Implications for Paleo-Pacific tectonics and magmatism in the Cretaceous. *Earth-Science Reviews* 220, 1-22. <https://doi.org/10.1016/j.earscirev.2021.103703>
23. **Kutyrev, A.**, Zelenski, M., Nekrylov, N., Savelyev, D., Kontonikas-Charos, A., Kamenetsky, V.S., 2021. Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano. *Frontiers in Earth Science* 9. <https://doi.org/10.3389/feart.2021.791465>
24. **Kutyrev, A.V.**, Sidorov, E.G., Kamenetsky, V.S., Chubarov, V.M., Chayka, I.F., Abersteiner, A., 2021. Platinum mineralization and geochemistry of the Matysken zoned Ural-Alaskan type complex and related placer (Far East Russia). *Ore Geology Reviews* 130, 1-17. <https://doi.org/10.1016/j.oregeorev.2020.103947>
25. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Agakhanov, A.A., Sandimirova, E.I., Vymazalová, A., Chubarov, V.M., Zolotarev, A.A., 2021. Kufahrite, PtPb, a new mineral from Ledyanoy Creek placer, Galmoenan ultramafic complex, Koryak Highlands, Russia. *Mineralogical Magazine* 85, 254-261. <https://doi.org/10.1180/mgm.2021.18>
26. Stepanov, S.Y., **Kutyrev, A.V.**, Lepekhina, E.N., Sharpenok, L.N., Antonov, A.V., Kutyreva, M.E., 2021. Age of the dike complex in the dunite "core" of the Kamenushinsky clinopyroxenite-dunite massif, Ural Platinum belt, Middle Urals. *Geochemistry International* 59, 559-576. <https://doi.org/10.1134/s0016702921060094>
27. Sidorov, E., **Kutyrev, A.**, Chubarov, V., Zhitova, E., 2021. Platinum mineralization of the Epilchik Ural-Alaskan type zoned complex (Far East Russia). *Mineralium Deposita* 56, 143-160. <https://doi.org/10.1007/s00126-020-01008-1>
28. Ismagilova, R.M., Zhitova, E.S., Krivovichev, S.V., Sergeeva, A.V., Nuzhdaev, A.A., Anikin, L.P., Krzhizhanovskaya, M.G., Nazarova, M.A., Kupchenko, A.N., Zolotarev, A.A., **Kutyrev, A.V.**, Bukhanova, D.S., Kuznetsov, R.A., Khanin, D.A., 2021. Phase evolution from volborthite,  $\text{Cu}_3(\text{V}_2\text{O}_7)(\text{OH})_2 \cdot 2\text{H}_2\text{O}$ , upon heat treatment. *Minerals* 11, 1312. <https://doi.org/10.3390/min11121312>
29. Saveliev, D.E., Shilovskikh, V.V., Sergeev, S.N., **Kutyrev, A.V.**, 2021. Chromian spinel neomineralisations and the microstructure of plastically deformed ophiolitic peridotites (Kraka massifs, Southern Urals, Russia). *Mineralogy and Petrology* 115, 411-430. <https://doi.org/10.1007/s00710-021-00748-w>
30. **Kutyrev, A.V.**, Kamenetsky, V.S., Sidorov, E.G., Abersteiner, A., Chubarov, V.M., 2020. Silicate inclusions in isoferroplatinum: Constraints on the origin of platinum mineralization in podiform chromitites. *Ore Geology Reviews* 119, 1-13. <https://doi.org/10.1016/j.oregeorev.2020.103367>
31. Chayka, I.F., Sobolev, A.V., Izokh, A.E., Batanova, V.G., Krashennnikov, S.P., Chervyakovskaya, M.V., Kontonikas-Charos, A., **Kutyrev, A.**, Lobastov, B., Chervyakovskiy, V., 2020. Fingerprints of Kamafugite-Like Magmas in Mesozoic Lamproites of the Aldan Shield: Evidence from Olivine and Olivine-Hosted Inclusions. *Minerals* 10, 1-30. <https://doi.org/10.3390/min10040337>
32. Stepanov, S.Y., Shagalov, E.S., Palamarchuk, R.S., **Kutyrev, A.V.**, Sharpenok, L.N., Nabiullin, F.M., Troshkina, A.N., 2020. Mineralogical and geochemical evidence for granite metasomatism in dikes in the north of the Berezovskoe Ore field (Middle Urals). *Russian Geology and Geophysics* 61, 809-827. <https://doi.org/10.15372/RGG2019176>
33. **Kutyrev, A.V.**, Zhirnova, T.S., 2019. Concentrically-zoned massifs of the Tamanvayam group (Koryak-Kamchatka Platiniferous Belt): structure, age, petrological and geochemical aspects. *Russian Journal of Pacific Geology* 13, 350-363. <https://doi.org/10.1134/S1819714019040031>

34. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Chubarov, V.M., Khanin, D.A., 2019. Origin of platinum-group mineral assemblages from placers in rivers draining from the Ural-Alaskan type Itchayvayamsky ultramafics, Far East Russia. *Canadian Mineralogist* 57, 91-104. <https://doi.org/10.3749/canmin.1800040>
35. **Kutyrev, A.V.**, Sidorov, E.G., Antonov, A.V., Chubarov, V.M., 2018. Platinum-group mineral assemblage of the Prizhimny Creek (Koryak Highland). *Russian Geology and Geophysics* 59, 935-944. <https://doi.org/10.1016/j.rgg.2018.07.014>

#### **Under review**

1. **Kutyrev, A.**, Maier, W., Karinen, T., Barnes, S.-J., Hayes, B., Savard, D., Virtanen, V., Booth, C., Muir, D. Petrogenesis of the Akanvaara layered intrusion, Finnish Lapland. *Journal of Petrology*, under review.
2. **Kutyrev, A.**, Bundtzen, T.K., Yakubovich, O.V., Chayka, I.F. Recognizing overlooked Ural-Alaskan-type complexes in ophiolitic belts: the Mount Hurst case, Alaska. *Mineralium Deposita*, under review.
3. Chayka, I., Hu, W.-J., **Kutyrev, A.**, Petrenko, D., Zhitova, L., Shevko, A., Gora, M., Kalugin, V., Shvedov, G., Izokh, A., Yudovskaya, M. Fluid-mediated PGE enrichment of magmatic low-sulfide ores as modelled for the Norilsk deposits. *Geology*, under review.

#### **Datasets and code**

1. **Kutyrev, A.V.**, Maier, W.D., Karinen, T., Barnes, S.J., Savard, D., 2025. Whole-rock geochemical dataset of the Akanvaara layered intrusion, Lapland, Finland. Dataset deposited with the Interdisciplinary Earth Data Alliance (IEDA). <https://doi.org/10.60520/IEDA/113569>
2. Filimonova, O., **Kutyrev, A.**, Wermeille, D., 2025. Iridium L<sub>3</sub>-edge XAS dataset of natural Ir-O phase, comprising IrO<sub>2</sub> [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.15650871>
3. Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Baykov, N.I., Barnes, S., Savard, D., Alexei, I.V., 2025. Major, trace, and platinum-group elements in the Late Cretaceous arc volcano-plutonic complex of the Tumrok Range, eastern Kamchatka, Version 1.0. Interdisciplinary Earth Data Alliance (IEDA). <https://doi.org/10.60520/IEDA/113578>
4. **Kutyrev, A.**, Bindeman, I., O'Neil, J., Rizo, H., 2024. Code for 1- and 2-stage oxygen fractionation and metamorphic dehydration modeling from "Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada". <https://doi.org/10.5281/zenodo.12795578>

#### **Conference presentations**

1. IMSG 2026, Barberton, South Africa. Oral presentation: "Mottled anorthosites and discordant ultramafic pipes in the MG4-UG1 interval of the Upper Critical Zone, Bushveld Complex".
2. Geocongress 2025, Bloemfontein, South Africa. Oral presentation: "Petrogenesis of the Akanvaara layered intrusion".
3. MDSG 2024, Edinburgh. Poster: "Assessing mass-balance of layered intrusions: the Akanvaara intrusion case study in Lapland, Finland".
4. Goldschmidt 2023, Lyon, France. Remote oral talk: "Early Earth's hydrosphere: insights from the hydrothermally altered submarine lavas of the Eoarchean Saglek-Hebron Complex, Canada".
5. 14th International Platinum Symposium 2023, Cardiff, UK. In-person oral presentation: "Origin of native osmium in metamorphosed peridotites: insights from multiphase inclusions (Adamsfield placer, Tasmania)".
6. Virtual Early Career Platinum Symposium 2022. Oral presentation: "Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano, Kamchatka".
7. EGU General Assembly 2021. Oral presentation: "Awaruite (Ni<sub>3</sub>Fe) as a platinum-group elements concentrator: Preliminary data".
8. SGA 2019 Conference, Glasgow, United Kingdom. In-person oral presentation: "Platinum mineralization in Ural-Alaskan complexes of Koryak Highlands (Far East Russia)".